Bowen Yu

Email: bowenyu@mit.edu | **Website**: bowenyu066.github.io
LinkedIn: www.linkedin.com/in/bowenyu066/ | **Telephone**: +1 (617) 685-9188

Education

Massachusetts Institute of Technology *(B.S., Physics & Artificial Intelligence and Decision Making)* — *Sep. 2024 — Jun. 2027*

Peking University *(Physics, transferred to MIT)* — *Sep. 2023 — Jul. 2024*

- **GPA**: 5.00/5.00 (*MIT*); 3.91/4.00 (*Peking Univ.*)
- *Selected Coursework*: Deep Learning, Natural Language Processing, Discrete Probability & Stochastic Processes, AI System Engineering, Computational Architecture, Quantum Physics I, II & III

Research & Industry Experiences

Research Assistant @ Prof. Yilun Du, Harvard University (*Cambridge, MA*) — *Sep. 2026 — Present*

- Ongoing research on how localized rewards shape emergent cooperation and hierarchical, efficient organization in multi-agent systems

Quantitative Research Intern @ Citadel Securities (*Miami, FL*) — *Jun. 2026 — Aug. 2026*

- ML-based systematic equities alpha research on factor return prediction

Research Assistant @ Prof. Mingda Li, MIT (*Cambridge, MA*) — *Jan. 2025 — Jun. 2026*

- Developed ML workflows for vibrational-spectra defect identification, grain boundary dynamics, and evaluation of generated crystal structures
- Co-first-authored four papers in ML-based defect engineering and valence-informed generative material discovery

Large Language Model Data Researcher @ ByteDance (*Beijing, China*) — *Aug. 2025 — Sep. 2025*

- Led the design of a benchmark suite of 500+ college-to-PhD-level physics problems for evaluating LLM reasoning
- Contributed to internal projects improving physical reasoning fidelity in frontier LLMs
- Co-developed a pipeline that achieves gold-medal–level performance on IPhO 2025 theoretical problems

AI Workload Deployment Intern @ Intel Corporation (*Shanghai, China*) — *Jun. 2025 — Jul. 2025*

- Authored internal tutorials on GPU memory layouts (CuTe layout, linear layout) for AI workloads
- Produced technical notes adopted by the CuTeDSL framework team as performance reference materials

Awards & Honors

International Physics Olympiad (IPhO) — *Jul. 2023*

Gold medal, Absolute Winner (**1st Overall Worldwide**)

Selected Publications

† → Equal contribution; * → Corresponding author

1. M. Cheng†,*, **Bowen Yu**†, C.-L. Fu, N. Andrejevic, M. Li*, *et al.* Probing Non-Equilibrium Grain Boundary Dynamics with XPCS and Domain-Adaptive Machine Learning. *arXiv preprint arXiv:2605.12194* (2026).
2. M. Cheng†,*, C.-L. Fu†, **Bowen Yu**†, M. Li*, *et al.* A foundation model for non-destructive defect identification from vibrational spectra. *Matter* **9** (2026).
3. M. Cheng†,*, W. Luo†, H. Tang†, **Bowen Yu**†, *et al.* Enhancing materials discovery with valence-constrained design in generative modeling. *Nature Computational Science* (2026).
4. M. Cheng†,*, Q. Wan†, **Bowen Yu**†, M. Li*, *et al.* Reinforcement learning-guided optimization of critical current in high-temperature superconductors. *arXiv preprint arXiv:2510.22424* (2025).

Selected Projects

Sparkie (HackMIT 2026) — *Sep. 2026*

- Co-built Sparkie, an AI teammate that joins live Zoom meetings like a real person, and *turns conversation directly into action*, without requiring users to switch to a separate chat window or re-explain any context
- Designed a dual-loop agent system that *listens continuously but speaks selectively*, while asynchronous tool-using agents execute tasks and bring results back into the ongoing conversation; Code / Project

Fireroad.ai (MIT CSAIL Agentic AI Hackathon 2026) — *Apr. 2026*

- Co-built a *domain-grounded* AI academic advisor that connects students' academic histories with MIT course data and degree requirements, turning fragmented information into personalized semester plans
- Built an agentic planning workflow that reasons over course offerings, checks degree requirements, and proposes actionable schedule edits
- Won the **Agents for MIT Track** at the MIT CSAIL Agentic AI Hackathon 2026; Code / Project